

Practice Midterm #2

Name: \_\_\_\_\_

ID: \_\_\_\_\_

Score: \_\_\_\_\_

| Problem | Pts. | Score |
|---------|------|-------|
| 1       | 20   |       |
| 2       | 20   |       |
| 3       | 20   |       |
| Total   | 60   |       |

I certify that this is my sole work and I haven't discussed any problem with other person.

Signature: \_\_\_\_\_

1. Please print your name and student ID number.
2. Sign on the above statement.
3. For full credit, you should show all of your work and explain your reasoning.

**GOOD LUCK!**

Symmetric & odd length

**Problem 1 (20 Pts) (a) (10 Pts)** Let  $h[n]$  be an FIR filter of GLP type I, i.e., the length  $L$  of the filter is an odd number and the impulse response shows the symmetry around the mid-point. Let  $g[n]$  be obtained through its  $z$ -transform satisfying

$$G(z) = -z^{-(L-1)}H(-z^{-1})$$

where  $H(z)$  denotes the  $z$ -transform of  $h[n]$ . Determine whether  $g[n]$  has GLP or not.

$$H(z) = \sum_{n=0}^{L-1} h[n] z^{-n}$$

$$H(-z^{-1}) = \sum_{n=0}^{L-1} h[n] (-z^{-1})^{-n}$$

$$-z^{-(L-1)} H(-z^{-1}) = \sum_{n=0}^{L-1} h[n] (-1)^{-n} z^n (-1) z^{-(L-1)}$$

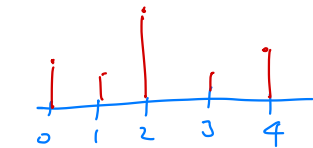
$$= \sum_{n=0}^{L-1} h[n] (-1)^{n+1} z^{-(L-1-n)}$$

$$m = L-1-n$$

$$= \sum_{m=0}^{L-1} h[L-1-m] (-1)^{L-m} z^{-m}$$

$$= \sum_{m=0}^{L-1} (-1)^{m-1} h[L-1-m] z^{-m}$$

$$g[m]$$

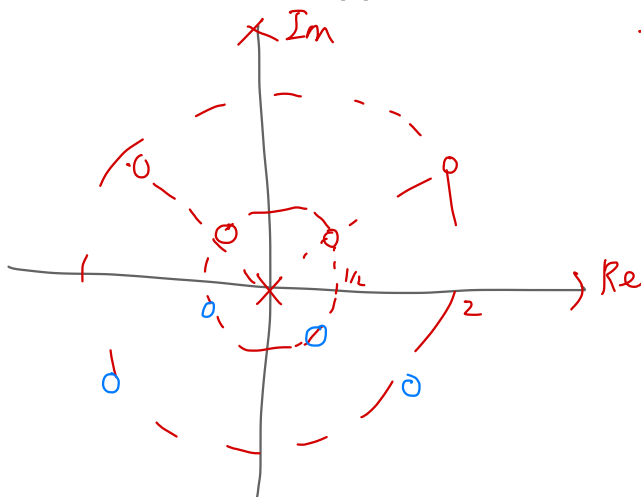


$$\begin{aligned} (-1)^{L-m} &= (-1)^L (-1)^{-m} \\ &= (-1) (-1)^m \\ &= (-1)^{m+1} \end{aligned}$$

**(b) (10 Pts)** Consider a discrete-time filter  $h[n]$  whose  $z$ -transform is given by

$$H(z) = z^{-4}(z - 0.5e^{j\pi/4})(z + 0.5e^{j\pi/4})(z - 2e^{j\pi/4})(z + 2e^{j\pi/4}).$$

Determine whether  $h[n]$  has GLP or not.



$$\begin{aligned} -0.5e^{j\pi/4} &= 0.5e^{j\pi}e^{j\pi/4} \\ &= 0.5e^{j(\pi+\pi/4)} \end{aligned}$$

**Problem 2 (20 Pts)** Suppose that we are given a continuous-time lowpass filter with frequency response  $H_c(j\Omega)$  such that

$$1 - \delta_p \leq |H_c(j\Omega)| \leq 1 + \delta_p, \quad |\Omega| \leq \Omega_p,$$

$$|H_c(j\Omega)| \leq \delta_s, \quad |\Omega| \geq \Omega_s.$$

A set of discrete-time lowpass filters is derived from  $H_c(s)$  via the bilinear transformation

$$H(z) = H_c(s) \Big|_{s = \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}}$$

parameterized by  $T > 0$ .

(a) (10 Pts) Show that under this transformation, the imaginary axis  $s = j\Omega$  in the  $s$ -plane is mapped to the unit circle  $z = e^{j\omega}$  in the  $z$ -plane. Derive the relationship between  $\omega$  and  $\Omega$ .

$$\frac{T}{2} s = \frac{z-1}{z+1} \quad \frac{T}{2} s (z+1) = z-1 \quad 1 + \frac{T}{2} j\Omega = re^{j\theta}$$

$$\frac{T}{2} s + 1 = (1 - \frac{T}{2} s) z \quad 1 - \frac{T}{2} j\Omega = re^{-j\theta}$$

$$z = \frac{1 + \frac{T}{2} s}{1 - \frac{T}{2} s}$$

$$T > 0$$

$$|z| = \left| \frac{1 + \frac{T}{2} j\Omega}{1 - \frac{T}{2} j\Omega} \right| = 1 \quad e^{j\omega} = \frac{1 + \frac{T}{2} j\Omega}{1 - \frac{T}{2} j\Omega}$$

$$\omega = 2\theta$$

$$= 2 \arctan\left(\frac{T\Omega}{2}\right)$$

(b) (10 Pts) Given a fixed  $\Omega_p$ , find the value of  $T$  such that the corresponding cutoff frequency for the discrete-time system is  $\omega_p = \pi/2$ .

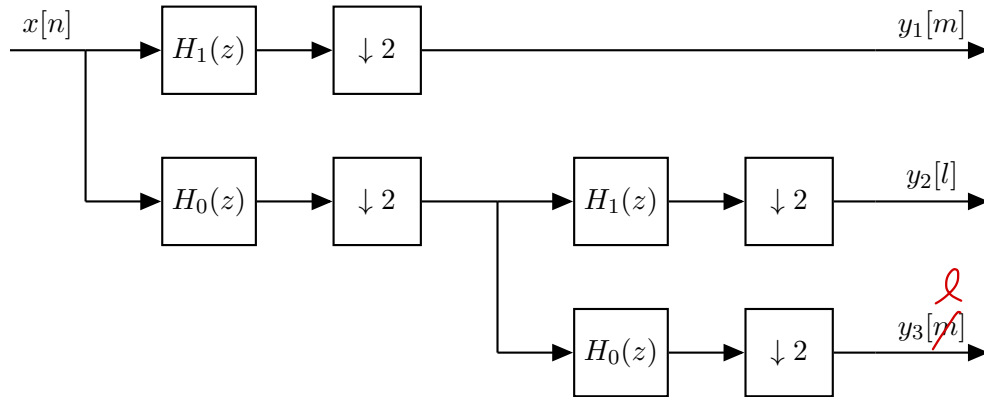
$$\omega_p = \frac{\pi}{2} = 2 \arctan\left(\frac{T\Omega}{2}\right)$$

$$\arctan\left(\frac{T\Omega}{2}\right) = \frac{\pi}{4}$$

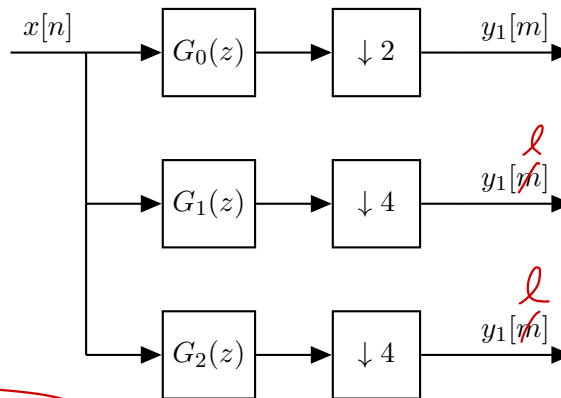
$$\frac{T\Omega}{2} = \tan\left(\frac{\pi}{4}\right) = 1$$

$$T = \frac{2}{\Omega_p}$$

**Problem 3 (20 Pts)** Consider the following filterbank system with  $H_0(z) = 1 + z^{-1}$  and  $H_1(z) = 1 - z^{-1}$ :



(a) (10 Pts) Find  $G_0(z)$ ,  $G_1(z)$ ,  $G_2(z)$ , expressed in terms of  $H_0(z)$  and  $H_1(z)$ , such that the following system produces the same outputs.



$$G_1(z) = \boxed{H_0(z)} \boxed{H_1(z^2)} \underbrace{\downarrow 2 \downarrow 2}_{\downarrow 4} \text{ — No block identity}$$

$$G_2(z) = \boxed{H_0(z) - H_0(z^2)} \underbrace{\downarrow 2 \downarrow 2}_{\downarrow 4}$$

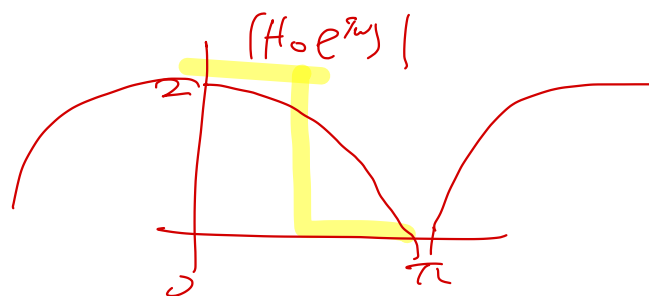
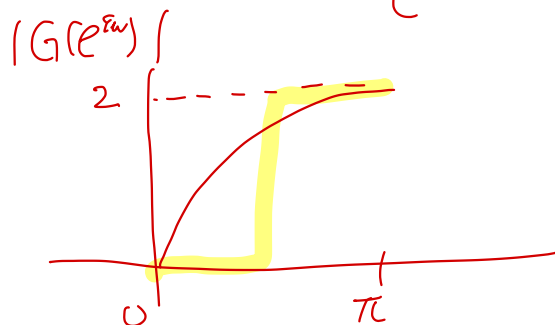
$$G_1(z) = H_0(z) H_1(z^2)$$

$$G_2(z) = H_0(z) H_0(z^2)$$

(b) (10 Pts) Sketch the magnitude responses of  $G_0(e^{j\omega})$ ,  $G_1(e^{j\omega})$ , and  $G_2(e^{j\omega})$ .

$$G_0(z) = H_1(z) = 1 - z^{-1}$$

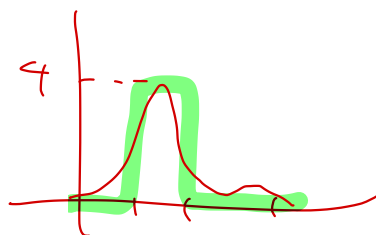
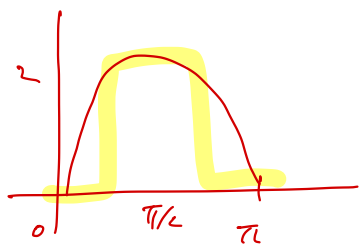
$$\begin{aligned} G_0(e^{j\omega}) &= 1 - e^{-j\omega} = e^{-j\frac{\omega}{2}} (e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}) \\ &= e^{-j\frac{\omega}{2}} 2j \sin\left(\frac{\omega}{2}\right) \\ &= e^{-j(\frac{\omega}{2} + \pi)} 2 \sin\left(\frac{\omega}{2}\right) \end{aligned}$$



$$G_1(e^{j\omega}) = H_0(z) H_1(z^2)$$

$$H_0(z) = 1 + z^{-1} \quad H_0(e^{j\omega}) = 2e^{j\frac{\omega}{2}} \cos\left(\frac{\omega}{2}\right)$$

$$\begin{aligned} |G_1(e^{j\omega})| &= |H_0(e^{j\omega})| |H_1(e^{j2\omega})| \\ &= 2 \cos\left(\frac{\omega}{2}\right) |2 \sin(\omega)| \end{aligned}$$



$$G_2(e^{j\omega}) = H_0(e^{j\omega}) H_0(e^{j2\omega})$$

